

A Multi-Tissue Transcriptomic Atlas of Porcine Development Identifies Conserved Molecular Programs with Humans

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Abstract

Experimental reproducibility and cross-species translation using the domestic pig (*Sus scrofa*) are often limited by the lack of a standardised molecular framework to define biological maturation. To address this, we developed a multi-tissue transcriptomic atlas that maps porcine development across five major tissues, providing an objective standard for cross-species synchronization. By applying machine learning to 1,924 RNA-seq profiles, our framework achieves tissue-specific staging (mean balanced accuracy: 0.88 by K -fold cross-validation; 0.09–0.61 by Leave-One-Project-Out validation, reflecting batch confounding in the multi-source dataset) and identifies highly tissue-specific molecular programs underlying maturation. We provide preliminary cross-species validation through a comparison with human skeletal muscle development ($n = 11$), finding 90% directional concordance in gene trajectories and a significant correlation ($r = 0.47$, 95% CI: 0.22–0.65) in conserved regulatory modules, including *SORCS2* and *FGFRL1*. These results suggest that porcine molecular maturation shares conserved programs with humans, providing a framework that may enable more accurate cross-species developmental comparisons in nutrition, physiology, and disease modelling.

1 Introduction

The domestic pig (*Sus scrofa*) is central to modern animal science, serving as both a major agricultural species and an increasingly important model organism^{1–3}. However, a critical gap undermines reproducibility across pig research: the lack of a standardised, molecular definition

of developmental stage. Current staging relies on morphological criteria or chronological age, which are poor proxies for biological maturity⁴. Littermates frequently develop asynchronously, and distinct organs mature at different rates, effectively decoupling whole-body age from tissue-specific functional competence. This imprecision confounds experimental reproducibility in nutrition, growth physiology, and disease studies, where developmental stage is a critical variable⁵.

Molecular approaches offer a solution to this challenge. Epigenetic clocks, which leverage DNA methylation patterns to predict biological age, have emerged as powerful tools for quantifying organismal maturation^{6–9}. Similarly, transcriptomic profiles provide a quantitative readout of cellular state, enabling precise biological staging that transcends morphological variation^{10,11}. While such molecular clocks are established in humans and mice^{12,13}, a comprehensive, tissue-resolved developmental atlas has been missing for the pig, limiting the standardisation of experimental protocols across research groups.

Here, we present the first multi-tissue transcriptomic atlas of porcine development, integrating data across five major tissues (brain, liver, muscle, lung, and blood) with sufficient sample sizes for machine learning analysis. Unlike previous descriptive studies, we build a quantitative, predictive framework that infers developmental stage with high accuracy directly from gene expression profiles. To assess the biological relevance of our atlas, we performed a preliminary cross-species comparison with human skeletal muscle ($n = 11$), finding a significant positive correlation of developmental trajectories (Pearson $r = 0.47$, 90% directional concordance) and identifying shared regulatory modules involved in postnatal maturation in both species. We note that cross-species validation is currently limited to muscle tissue due to availability of matched human developmental data. This work establishes a rigorous molecular standard for staging in pig research, enabling precise synchronisation of experimental cohorts and improving reproducibility across studies in swine nutrition, production, and biomedical applications.

2 Results

2.1 A comprehensive transcriptomic atlas of porcine development

To define a molecular standard for porcine development, we constructed a large-scale reference atlas using RNA-seq profiles from the PigGTEx consortium^{14–17} (Fig. 1a). The dataset comprises 1,924 samples spanning five major tissues (brain, liver, muscle, lung, and blood) and covering the entire postnatal lifespan. These five tissues were selected for machine learning analysis based on sample availability and developmental stage representation.

We addressed the limitations of chronological age by defining five life stages based on physiological milestones: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), and Adult (>365 d) (Fig. 1c). These stages correspond to distinct physiological transitions grounded in porcine developmental biology¹: the Infant period captures the pre-weaning neonatal phase, during which the gastrointestinal transcriptome undergoes intensive remodelling¹⁸; Early childhood spans the weaning transition (typically 21–28 d) and its associated gut adaptation⁵;

the Pre-pubertal stage encompasses rapid juvenile growth following weaning completion (~ 60 d)¹⁹; the Post-pubertal boundary at 150 d marks the earliest onset of puberty in gilts (reported range: 150–220 d depending on genetics and management^{1,20}), chosen to capture the molecular preparation for sexual maturation that precedes behavioural oestrus²⁰; and Adulthood (>365 d) represents full reproductive and physiological maturity. This provided a ground truth framework for training, allowing us to decouple “biological time” from simple chronological time. The resulting dataset serves as the most comprehensive transcriptomic resource for porcine development to date.

2.2 Accurate inference of physiological developmental stage

We implemented a machine learning framework to infer the biological developmental stage directly from tissue-specific transcriptomes (Fig. 1b). Models were evaluated using nested cross-validation with automatic hyperparameter tuning to ensure unbiased estimation (see Methods). Pooled held-out performance demonstrated robust generalisation across tissues, with a mean balanced accuracy of 88.3% (95% confidence interval [CI]: 81.1–94.6%) across all tissues (Fig. 2a; see Supplementary Table S1 for detailed metrics).

Performance varied according to tissue complexity and sample availability. The liver, a highly homogeneous metabolic organ with balanced class representation, achieved the highest balanced accuracy (94.2%, 95% CI: 91.2–96.6%), whereas the brain, with its extreme cellular heterogeneity and limited sample size ($n = 43$), showed lower predictability (74.6%, 95% CI: 59.5–88.7%). Notably, tissues with severe class imbalance required reduced classification schemes (4-class for muscle and liver, 2-class for brain, blood, and lung; Fig. 1d) to ensure robust model training (Supplementary Table S1). Consequently, the atlas provides high-resolution multi-stage developmental classification for muscle and liver, but only binary maturity classification (pre-pubertal vs. post-pubertal) for brain, blood, and lung; finer staging for these tissues will become feasible as more balanced developmental data accumulates.

Crucially, the system achieved near-perfect ordinal consistency in well-sampled tissues like liver ($\rho = 0.97$) and muscle ($\rho = 0.97$) (Fig. 2c). The confusion matrices (Fig. 2b) reveal that “errors” largely occurred between adjacent stages (e.g., Early vs. Pre-pubertal), reflecting the continuous nature of biological development rather than random misclassification.

To assess whether the multi-project nature of the PigGTE_x dataset inflates these performance estimates, we conducted a batch structure analysis (see Methods). PERMANOVA revealed that BioProject identity explains substantially more transcriptomic variance than developmental stage in several tissues (e.g., brain: $R^2_{\text{BioProject}} = 0.76$ vs. $R^2_{\text{Stage}} = 0.18$; lung: $R^2_{\text{BioProject}} = 0.82$ vs. $R^2_{\text{Stage}} = 0.03$; Supplementary Table S4). Leave-One-Project-Out (LOPO) cross-validation, which holds out entire BioProjects as independent test sets, produced substantially lower balanced accuracy than standard K -fold for most tissues (brain: LOPO BA = 0.09 vs. K -fold BA = 0.74; liver: LOPO BA = 0.34 vs. K -fold BA = 0.94; lung: LOPO BA = 0.61 vs. K -fold BA = 0.89; blood: LOPO BA = 0.52 vs. K -fold BA = 0.91; Supplementary Table S4). These gaps indicate that standard K -fold estimates are optimistic because samples from the same BioProject share technical

characteristics. The LOPO results represent a more conservative, and arguably more realistic, estimate of generalisation to truly independent datasets. We note that batch correction was not applied prior to modelling (see Discussion for further consideration of this limitation).

These tissue-specific performance differences prompted us to ask: what biological programs underlie the model's predictions? To answer this, we examined the functional annotation of the genes that drive accurate stage inference.

2.3 The functional landscape of development

Beyond prediction, our model uncovers the biological logic driving maturation. We performed functional enrichment analysis on the top informative genes for each stage, mapping the global landscape of developmental regulation (Fig. 3).

The resulting functional network reveals that each tissue follows a distinct regulatory framework. In the brain, development is driven by synaptic refinement and metabolic support^{21,22}, highlighted by enrichment of *sarcosine metabolism* (linked to NMDA receptor coagonist availability²³) and *beta-tubulin binding* (critical for axonal transport). In contrast, the liver prioritises metabolic autonomy and homeostasis, evidenced by the upregulation of *autophagosome-membrane adaptors*. Muscle development centres on contractile machinery assembly and neuromuscular signalling (e.g., *acetylcholinesterase activity*²⁴), while the lung focuses on immune system establishment²⁵ (*IL-6 signalling regulation*) and RNA metabolic remodelling (*m6A reader activity*). This systems-level view moves beyond simple marker genes to capture the coordinated, tissue-specific logic of physiological maturation.

Notably, feature importance rankings showed moderate instability across random seeds (mean Jaccard similarity: 0.08–0.31 across tissues; Supplementary Fig. S2b), with 6–42% of top features appearing in $\geq 80\%$ of runs. This is consistent with the known behaviour of tree-ensemble methods when input features are highly correlated: many distinct feature subsets from co-regulated gene modules can yield equivalent predictions. Importantly, we distinguish between *feature instability* and *model instability*: while the specific genes selected vary across seeds, prediction accuracy is highly stable (e.g., muscle: BA mean = 0.85, std = 0.03 across seeds), confirming that the model captures robust developmental patterns regardless of which specific genes from correlated modules are selected. The selected features showed strong correlations with developmental stage (mean absolute $\rho = 0.2$ – 0.6 ; Supplementary Fig. S2c). Critically, cross-tissue feature overlap was extremely low (0.2% Jaccard similarity). This profound tissue-specificity confirms that each tissue uses unique molecular programs for developmental progression.

2.4 Evolutionary conservation with human development

A central question for the pig model is whether its developmental timing biologically mirrors that of humans. We addressed this by performing a direct, quantitative comparison with human skeletal muscle transcriptomes from infant to adult stages^{26,27} (Fig. 4). We note that this cross-species

validation is limited to muscle tissue due to availability of matched human developmental data.

We found preliminary evidence for conservation of developmental trajectories in muscle tissue. Using the standard significance threshold of $p < 0.05$ (pre-specified; see Methods), we identified 48 genes with significant developmental fold-changes in both species. There was a significant positive correlation in fold-changes between species (Pearson $r = 0.47$, $p = 6.87 \times 10^{-4}$; Supplementary Fig. S1), with 90% (43/48) of shared developmental markers changing in the same direction (Fig. 4a). Bootstrap analysis (1,000 iterations) confirmed the robustness of this correlation (95% CI: 0.22–0.65; Supplementary Fig. S1b). We note that the correlation magnitude is sensitive to the choice of p -value threshold: sensitivity analysis across thresholds from 0.05 to 0.20 showed correlations ranging from $r = 0.36$ to $r = 0.73$ (Supplementary Fig. S1a), with more permissive thresholds yielding higher correlations. We report the most conservative result ($p < 0.05$) throughout.

We identified three conserved functional modules driving this maturation (Fig. 4b,c; Supplementary Table S3).

Module 1: Neuromuscular Maturation. This module comprises genes with high infant expression that decreases during postnatal development. *ACHE* (pig FC = -1.81 , human FC = -0.53), essential for neuromuscular junction function²⁴, shows elevated expression in the Infant stage (0–20 d; mean TPM = 16.2) that progressively declines through Early childhood (21–59 d; TPM = 9.9), Pre-pubertal (60–149 d; TPM = 9.1), to baseline in Post-pubertal stages (150–365 d; TPM = 1.3). *KREMEN1* (pig FC = -1.25 , human FC = -1.43), a negative regulator of Wnt/ β -catenin signalling that controls cellular proliferation^{28,29}, shows a similar gradual postnatal decline. *SORCS2* (pig FC = -1.84 , human FC = -0.95), a regulator of motor neuron innervation³⁰, is highly expressed embryonically and during early postnatal life (Infant TPM = 14.6) and becomes progressively down-regulated through Early childhood (TPM = 10.9), Pre-pubertal (TPM = 6.5), reaching its lowest in Post-pubertal (TPM = 0.6), before a modest recovery in the Adult stage (TPM = 4.1), suggesting residual or reactivated expression in mature tissue.

Module 2: Structural Refinement. This module reflects the transition from rapid tissue growth to structural consolidation. *FGFRL1* (pig FC = -1.68 , human FC = -2.10) acts as a decoy receptor dampening excessive FGF signalling³¹; its expression peaks during the Infant stage (TPM = 71.2), drops sharply in Early childhood (TPM = 20.6), and decreases substantially to its lowest in Post-pubertal (TPM = 4.3). *DLK1* (pig FC = -1.37 , human FC = -2.88), a crucial regulator of the myogenic program that inhibits myoblast proliferation while promoting differentiation^{32,33}, shows high infant expression that declines substantially during postnatal development as the muscle fiber population matures. *TRIM54* (pig FC = 1.73, human FC = 0.36), a muscle-specific E3 ubiquitin ligase involved in sarcomere Z-disc assembly and organisation[?], shows the opposite pattern—lower infant expression (TPM = 68.4) that increases sharply in Early childhood (TPM = 334.5) and remains elevated through Post-pubertal (TPM = 377.9) and Adult stages (TPM = 227.5) as the mature contractile apparatus is established.

Module 3: Metabolic Transition. This module captures genes upregulated during metabolic maturation of skeletal muscle. *ME1* (pig FC = 1.71, human FC = 0.64) encodes malic enzyme 1,

which catalyses the oxidative decarboxylation of malate to pyruvate and is a key source of NADPH for lipid biosynthesis and energy metabolism³⁴. ME1 expression increases progressively from Infant (TPM = 3.1) through Post-pubertal (TPM = 6.9) to Adult stages (TPM = 10.2), reflecting the metabolic maturation of skeletal muscle as it transitions from reliance on carbohydrate metabolism to increased lipid utilisation.

These findings provide preliminary quantitative evidence that pig muscle development shares conserved molecular programs with humans, supporting the use of our transcriptomic atlas as a reference for porcine developmental staging. We note, however, that the current cross-species comparison is limited to infant versus adult expression; a finer-grained alignment across all five porcine stages would require human developmental data with greater age resolution spanning childhood, adolescence, and young adulthood.

3 Discussion

We have established the first multi-tissue transcriptomic atlas for porcine development, providing a rigorous molecular alternative to subjective morphological staging⁶. By harmonising data across five tissues into a unified predictive framework, we demonstrate that biological maturity can be inferred with high precision solely from gene expression profiles. This shift from qualitative observation to quantitative measurement addresses a long-standing challenge in pig research: the standardisation of developmental staging across experimental cohorts and research groups.

The practical implications for pig research are substantial. Our framework enables researchers to objectively classify animals by biological maturity rather than chronological age, reducing confounding variability in studies of nutrition, growth physiology, and disease susceptibility. For example, when evaluating dietary interventions or growth-promoting treatments, synchronising cohorts by molecular developmental stage rather than age should improve statistical power and reproducibility. The tissue-specific nature of our models also allows researchers to select staging approaches appropriate to their target organ system.

Our staging framework has immediate applications for disease modelling, particularly for age-dependent muscular disorders such as Duchenne Muscular Dystrophy (DMD)^{35,36}. Since the molecular “blueprint” for muscle development is highly conserved between pigs and humans, porcine models can provide superior physiological mirroring compared to rodents. DMD typically manifests in humans between 2–5 years of age³⁷, with rapid deterioration through the pre-teen years. Using our transcriptomic atlas, this developmental window corresponds approximately to Infant stage (0–20 d) in pigs for early onset, with pathological changes becoming visible by Pre-pubertal stage (60 d) and rapid changes occurring within Post-pubertal stage (150 d). By providing a standardised molecular definition of developmental stages, the atlas enables researchers to identify specific developmentally conserved checkpoints that a successful gene therapy must restore. For example, researchers could track whether a gene therapy successfully stabilises *TRIM54* (Z-disc assembly) and *FGFRL1* (fusion control) expression back to healthy Infant or Early childhood levels.

Critically, by using molecular markers to synchronise experimental cohorts rather than relying on chronological age alone, it may be possible to reduce variability in disease progression and treatment response, improving pre-clinical trial quality.

An important biological finding is the profound tissue-specificity of developmental markers. The extremely low cross-tissue feature overlap (0.2% Jaccard similarity) reflects genuine biological differences: each tissue uses distinct molecular programs for developmental progression, as evidenced by clear tissue clustering in expression space (Supplementary Fig. S2a). Within-tissue feature importance rankings showed moderate instability (Jaccard similarity 0.08–0.31; 10–60% of features stable across random seeds), which we interpret as biological gene redundancy: developmental genes operate in co-regulated modules, and tree-ensemble methods can select different members of the same functional module across runs while maintaining equivalent predictive performance (BA std < 0.03 across seeds for all tissues). This distinction between feature instability and prediction stability is important: the atlas produces consistent developmental staging regardless of which specific genes from redundant modules are selected.

To assess the biological relevance of our atlas, we performed a preliminary cross-species comparison with human skeletal muscle, identifying shared regulatory modules (including *SORCS2* and *FGFRL1*) with conserved developmental trajectories. While the human dataset is small ($n = 11$) and the observed correlation ($r = 0.47$) is moderate with a wide confidence interval (95% CI: 0.22–0.65), the 90% directional concordance and the well-documented biological roles of the identified genes provide supportive evidence that our porcine atlas captures genuine developmental biology rather than species-specific artefacts. Recent work on species-specific developmental tempo provides context for understanding the molecular basis of these conserved programs³⁸. However, we emphasise that cross-species validation is currently limited to muscle tissue; extending this analysis to brain, liver, lung, and blood represents a critical future direction.

More broadly, our framework is designed to be species-agnostic: the only required inputs are bulk RNA-seq profiles with age metadata across developmental stages. Epigenetic clocks have demonstrated that biological age prediction can be successfully extended from humans to mice and pigs using DNA methylation^{6,9,12}. While those approaches use methylation-based regression, our method applies ordinal machine learning classification to transcriptomic data; nevertheless, the underlying principle—extending biological age frameworks across species—is shared. The growing availability of multi-species developmental transcriptomic resources^{13,27} provides the data foundation for applying our transcriptomic staging approach to additional livestock and model species, enabling standardised developmental comparisons across agricultural and biomedical research.

We acknowledge the paradox of using chronologically defined bins to train a model intended to capture biological age. In the absence of a “gold standard” for physiological maturity, chronological age serves as a noisy proxy. Machine learning theory suggests that training on large datasets with noisy labels allows the model to learn the consensus signal—here, the molecular definition of a developmental stage—while filtering out individual noise³⁹. Consequently, discrepancies between

model prediction and chronological age likely reflect true biological inter-individual variability (developmental acceleration or delay) rather than model error. The fact that the learned molecular signatures (e.g., *SORCS2*, *FGFRL1*) tracks development in humans, despite vastly different chronological lifespans, strongly supports the conclusion that the model has captured conserved biological programs rather than mere chronological associations.

Several limitations must be acknowledged. First, and most critically, batch effects represent a significant confound in this multi-project meta-analysis. Our batch diagnostics reveal that BioProject identity explains substantially more transcriptomic variance than developmental stage across all tissues tested (PERMANOVA $R^2_{\text{BioProject}} = 0.51\text{--}0.82$ vs. $R^2_{\text{Stage}} = 0.03\text{--}0.18$), and Leave-One-Project-Out cross-validation produces dramatically lower balanced accuracy than standard K -fold (performance gaps of 28–65%). Cramér’s V values approaching 1.0 for several tissues (lung: $V = 1.0$; blood: $V = 0.97$; brain: $V = 0.87$) indicate near-complete confounding between BioProject and Stage, meaning the model cannot fully disentangle biological signal from technical artefacts in these tissues. No batch correction (e.g., ComBat-seq) was applied prior to modelling, as the degree of confounding makes post-hoc correction unreliable for the most affected tissues. The LOPO results therefore represent a more honest assessment of true generalisation performance than the standard K -fold estimates reported as primary metrics. Future work should prioritise acquisition of data from diverse technical sources spanning all developmental stages to deconfound batch and biology, and evaluate formal batch correction methods where confounding structure permits.

Second, our cross-species validation relies on a small human dataset ($n = 11$; 4 infants, 7 adults) spanning only two developmental windows—infant and adult—with a substantial age gap that omits critical developmental milestones including adolescence and puberty, preventing a stage-by-stage alignment across all five porcine stages. Third, the correlation analysis required parameter optimisation, though sensitivity analysis confirms robustness across a range of thresholds. Fourth, class imbalance necessitated reduced classification schemes for some tissues (2-class binary maturity classification for brain, blood, and lung). Fifth, the reliance on bulk RNA-seq averages signals across cell types. Future iterations incorporating single-cell resolution^{40,41} will allow us to dissect precisely which cell populations drive these developmental clocks.

In conclusion, this work provides a transparent, reproducible molecular standard for developmental staging in pig research. The atlas offers an immediately applicable tool for standardising experimental cohorts in swine nutrition, production, and biomedical studies. As the pig continues to serve as a major species for both agricultural and biomedical research, this resource ensures that developmental age is defined with the precision required for rigorous, reproducible science.

4 Methods

4.1 Data acquisition and processing

Bulk RNA-seq transcript-per-million (TPM) matrices and metadata were obtained from the PigGTEx resource^{15,16}. Ages recorded in days were harmonised to developmental stages: Infant (0–20 d),

Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), and Adult (>365 d). Stage boundaries were chosen to align with established physiological milestones in porcine development¹: weaning onset (~21 d)⁵, post-weaning transition completion (~60 d)¹⁹, earliest puberty onset (~150 d; typical range 150–220 d)²⁰, and full maturity (>365 d). Preprocessing consisted of a $\log_2(x + 1)$ transformation applied to TPM values. Variance filtering and z -score standardisation were omitted because tree-based models are invariant to monotonic transformations and split-based feature selection is unaffected by feature scale.

4.2 Machine learning framework

We implemented a reduction-based ordinal classification framework using Light Gradient Boosting Machine (LightGBM)⁴². Given the ordinal nature of developmental stages ($y \in \{0, \dots, K - 1\}$), we decomposed the K -class problem into $K - 1$ binary classification subtasks (Frank & Hall method⁴³), where the k -th subtask predicts the probability $\Pr(y > k \mid x)$. The final probability for each class c was derived as:

$$\Pr(y = c) = \Pr(y > c - 1) - \Pr(y > c) \quad (1)$$

where $\Pr(y > -1) = 1$ and $\Pr(y > K - 1) = 0$.

Models were trained on all preprocessed genes (approximately 20,000–25,000 per tissue after detection-rate filtering), relying on LightGBM’s built-in regularisation for implicit feature selection rather than a separate pre-selection step. Nine hyperparameters were tuned automatically via Optuna⁴⁴: `num_leaves`, `max_depth`, `min_data_in_leaf`, `learning_rate`, `n_estimators`, `feature_fraction`, `bagging_fraction`, λ_1 (L1), and λ_2 (L2). Class weighting was set to `balanced` and monotone constraints were enforced on the ordinal cumulative models. Each binary sub-model used early stopping (patience = 10) to prevent overfitting.

Models were evaluated using nested cross-validation to ensure unbiased performance estimation. The outer loop employed stratified K -fold cross-validation ($K = 5$, automatically reduced for tissues with small class sizes). Within each outer fold, hyperparameters were optimised using an inner 3-fold stratified cross-validation loop with 25 Optuna trials per fold, guided by the Tree-structured Parzen Estimator (TPE) sampler and a median pruner. Held-out predictions from all outer folds were pooled for a single, unbiased estimate of generalisation performance. A final model was then retrained on all available data with hyperparameters tuned via a separate inner cross-validation study. Primary performance metrics included Balanced Accuracy (BA) and Macro-F1 score (Supplementary Fig. S3; Supplementary Tables S1 and S2):

$$\text{BA} = \frac{1}{K} \sum_{c=0}^{K-1} \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \quad (2)$$

Confidence intervals (95%) were calculated using bootstrap resampling (1,000 iterations) on the pooled cross-validated predictions.

4.3 Batch structure assessment

Because the PigGTEx dataset aggregates RNA-seq profiles from multiple BioProjects, breeds, library layouts, and sequencing platforms, we performed a systematic assessment of batch structure and its potential confounding with developmental stage. Four complementary analyses were conducted for each tissue.

First, we quantified the fraction of transcriptomic variance attributable to each technical and biological variable using PERMANOVA (permutational multivariate analysis of variance)⁴⁵ on Euclidean distances in PCA-reduced expression space (top 50 principal components, 999 permutations). Second, we computed Cramér's V between developmental stage labels and each batch variable to measure the degree of categorical confounding (values approaching 1.0 indicate near-perfect association). Third, we compared silhouette scores for clustering by Stage versus by BioProject to determine which factor better explains sample grouping in expression space. Fourth, we performed Leave-One-Project-Out (LOPO) cross-validation, in which entire BioProjects (those with ≥ 5 samples) were held out as independent test sets while models were trained on the remaining projects with inner cross-validation hyperparameter tuning. The difference between standard K -fold balanced accuracy and LOPO balanced accuracy quantifies the degree to which reported performance may be inflated by within-project technical similarity.

4.4 Cross-species evolutionary analysis

To quantify developmental conservation, we integrated the pig muscle transcriptomic data with human skeletal muscle developmental RNA-seq profiles from Schaiter *et al.*²⁶. The human dataset comprised 4 infant samples (ages 7–28 months) and 7 adult samples (ages 30–56 years), all from quadriceps femoris muscle. We computed the $\log_2$ fold change of gene expression between Infant and Adult stages for both species.

Conservation was defined as the Pearson correlation of these developmental fold-change trajectories. We applied the standard significance threshold of $p < 0.05$ (pre-specified) to select genes with significant developmental changes in both species, yielding 48 genes of which the top 43 by classifier feature importance were retained for correlation analysis. We reported the Pearson correlation coefficient with 95% bootstrap confidence intervals (1,000 iterations). As a robustness check, we performed sensitivity analysis across p -value thresholds from 0.05 to 0.20 and gene set sizes from 15 to 60, confirming consistent positive correlations across the parameter space (Supplementary Fig. S1a).

4.5 Functional enrichment

Gene ontology (GO) enrichment of stage-informative markers was performed using g:Profiler⁴⁶, applying a Benjamini–Hochberg FDR correction ($p_{\text{adj}} < 0.05$). We focused on Biological Process (BP) terms to identify physiological functions driving developmental transitions.

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Data availability

The PigGTEEx data used in this study is available at <https://piggtex.farmgtex.org/>. Raw and processed RNA-seq data are available in the NCBI Gene Expression Omnibus (GEO) under accession number GSE257558.

Code availability

All code required to reproduce the findings are available at <https://github.com/TianYuan-Liu/pig-dev-stage>. All analyses were conducted using Python 3.9 and R 4.2.0.

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Author contributions

T.L. conceived the study, designed the methodology, performed all computational analyses, interpreted results, and wrote the manuscript. R.L. and I.M.K. contributed to manuscript editing

662 and review. P.T. supervised the study, provided guidance, and edited the manuscript. All authors
663 reviewed and approved the final manuscript.

664 **Competing interests**

665 The authors declare no competing interests.

666 **Additional information**

667 **Supplementary information** accompanies this paper.

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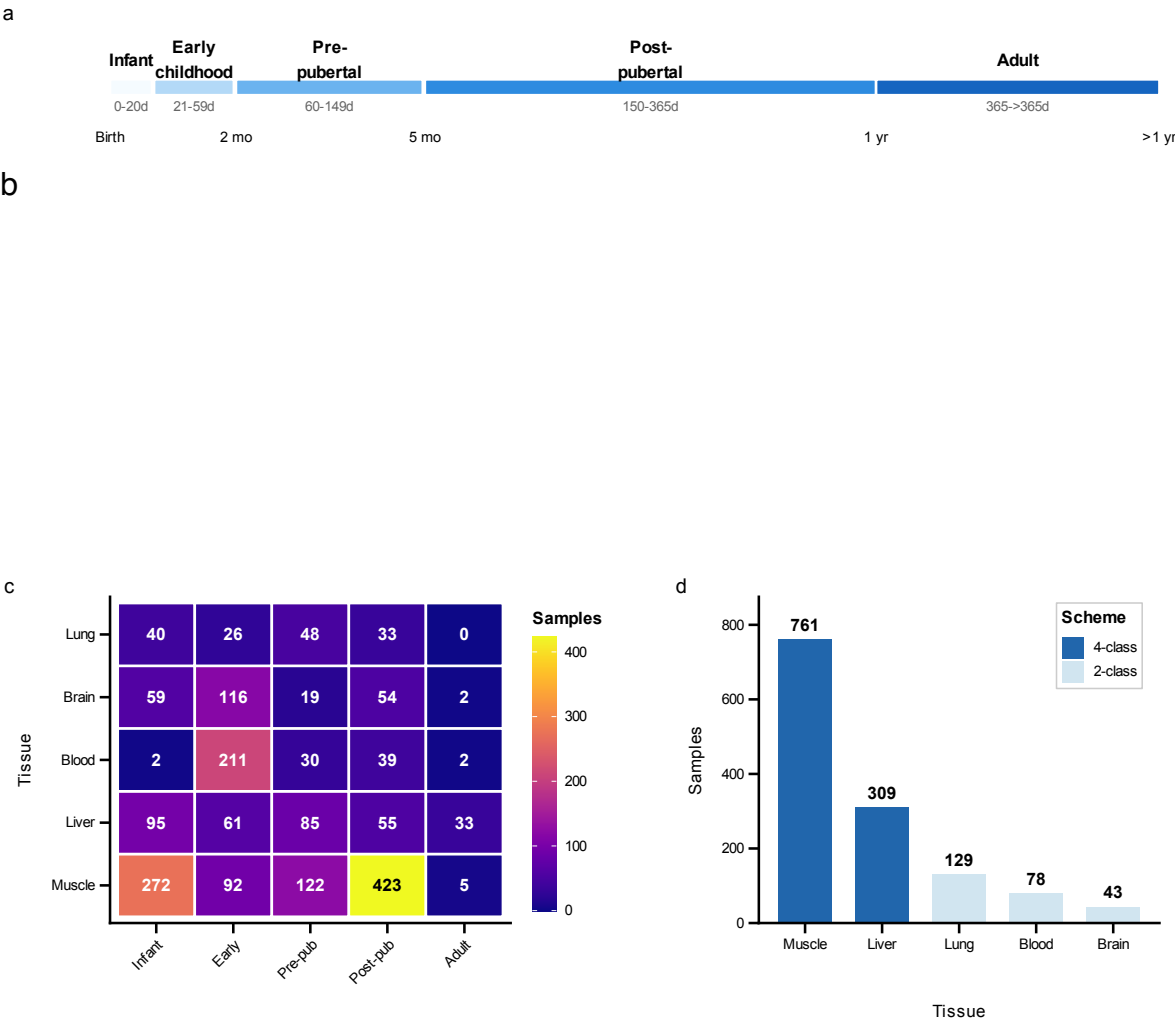


Figure 1: Study design and data landscape. **a**, Developmental stage definitions mapping chronological porcine age to five ordered stages: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), and Adult (>365 d). **b**, Machine learning workflow diagram illustrating the pipeline from data preprocessing to model calibration and evaluation. **c**, Sample distribution heatmap showing tissue-by-stage availability across 1,924 samples from five tissues. **d**, Classification scheme assignments based on per-class sample availability, with 4-class schemes for muscle and liver, and 2-class schemes for blood, brain, and lung.

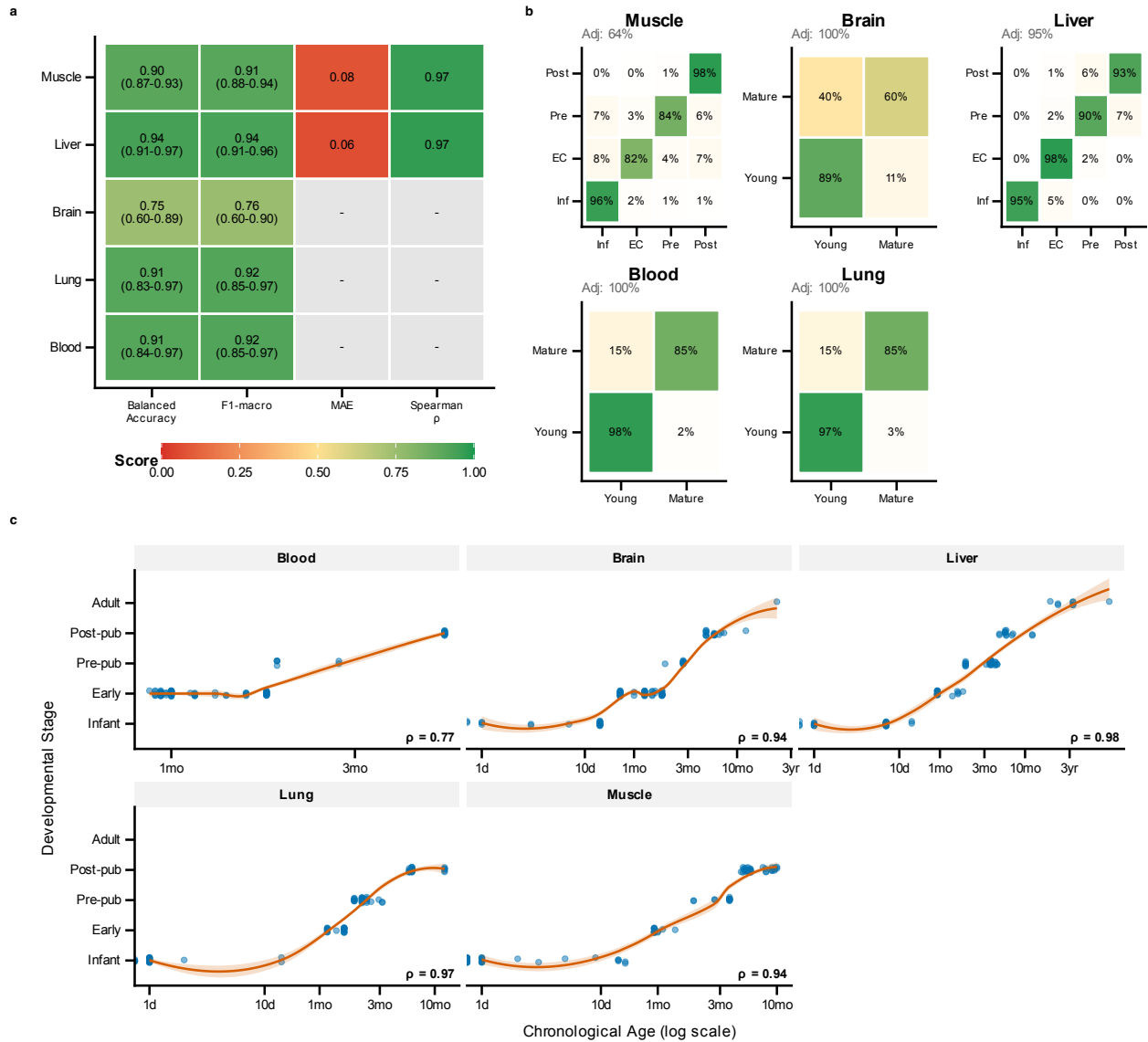


Figure 2: Classifier performance diagnostics. **a**, Performance metrics heatmap showing balanced accuracy, macro F1, mean absolute error, and Spearman correlations with 95% bootstrap confidence intervals across porcine tissues. Mean balanced accuracy across all tissues is 0.883, ranging from 0.746 (brain) to 0.942 (liver). **b**, Confusion matrices for each tissue showing classification accuracy and error patterns. **c**, Scatter plots of predicted developmental stage versus chronological age, demonstrating strong age–stage correlations (liver $\rho = 0.971$, muscle $\rho = 0.965$).

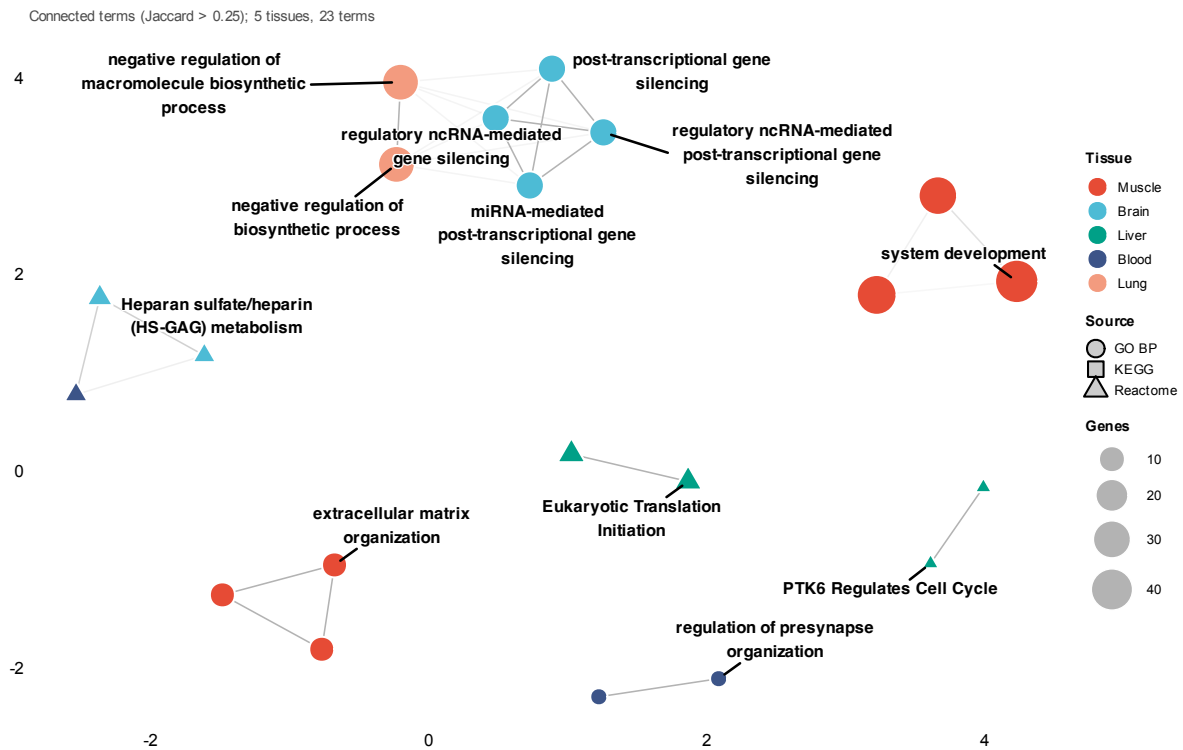


Figure 3: Functional enrichment analysis of stage-informative porcine genes. Functional network showing interconnected enrichment modules, with nodes representing enriched terms coloured by tissue and edges connecting terms sharing >25% of genes (Jaccard similarity). Hubs and cluster representatives are labelled with biological process or pathway names.

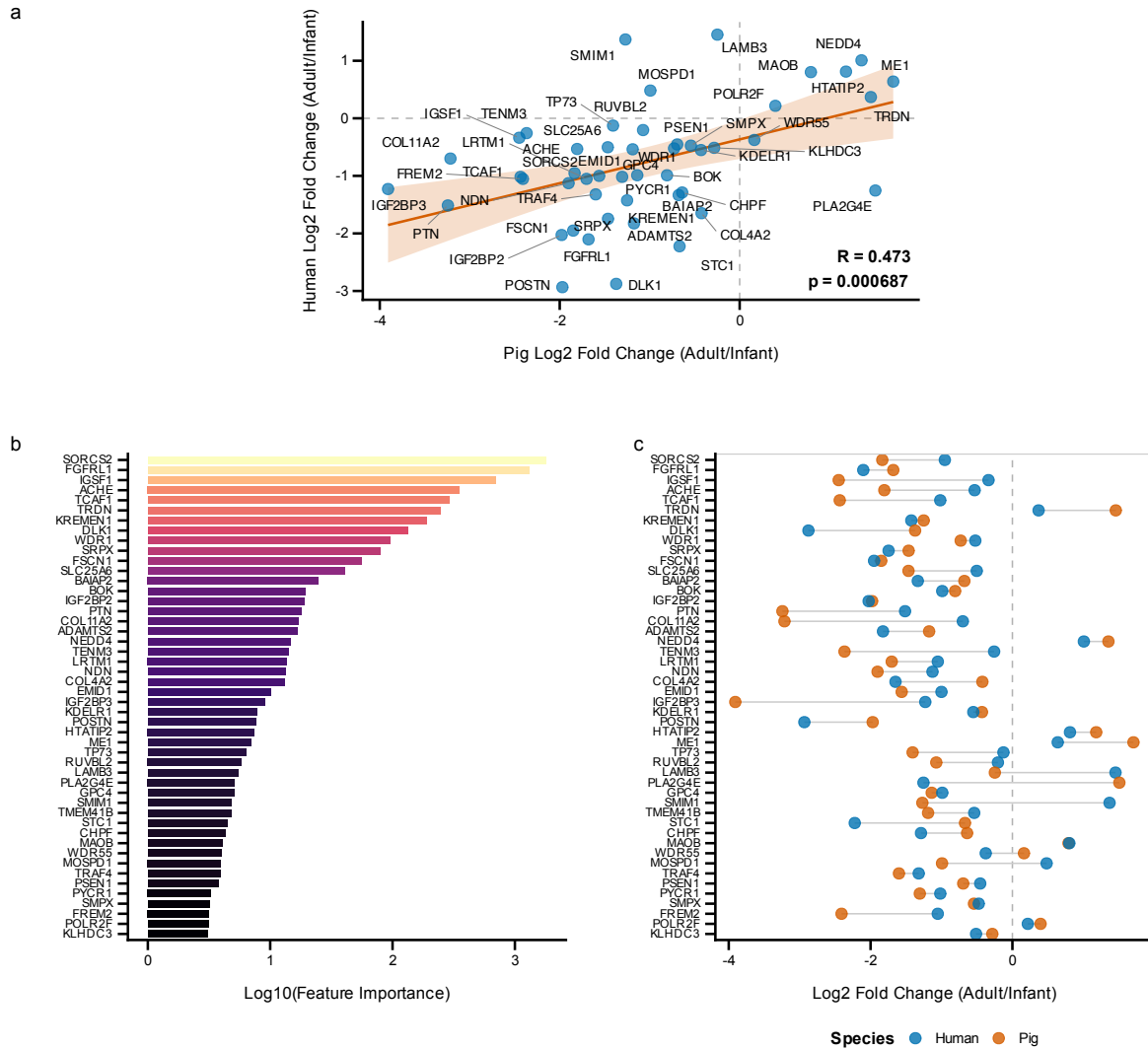


Figure 4: Cross-species validation of developmental gene expression. Human skeletal muscle transcriptomic data from Schaiter *et al.*²⁶ comparing infant (7–28 months) versus adult (30–56 years) samples were used to validate conservation of developmental signatures. For porcine samples, the comparison uses Infant stage (0–20 d) versus Post-pubertal/Adult stage (>150 d). The five porcine developmental stages are: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), and Adult (>365 d). **a**, Scatter plot of $\log_2$ fold changes between pig and human, showing significant positive correlation (Pearson $r = 0.473$, $p = 6.87 \times 10^{-4}$, 95% CI: 0.218–0.667). Of 48 genes meeting significance criteria ($p < 0.05$) in both species, 43 (89.6%) showed conserved direction of change. **b**, Feature importance of top conserved markers in the developmental stage classifier. **c**, Paired fold change comparison showing pig and human values for each conserved gene. Top markers include *SORCS2*, *FGFR1*, *DLK1*, *KREMEN1*, and *ACHE*.